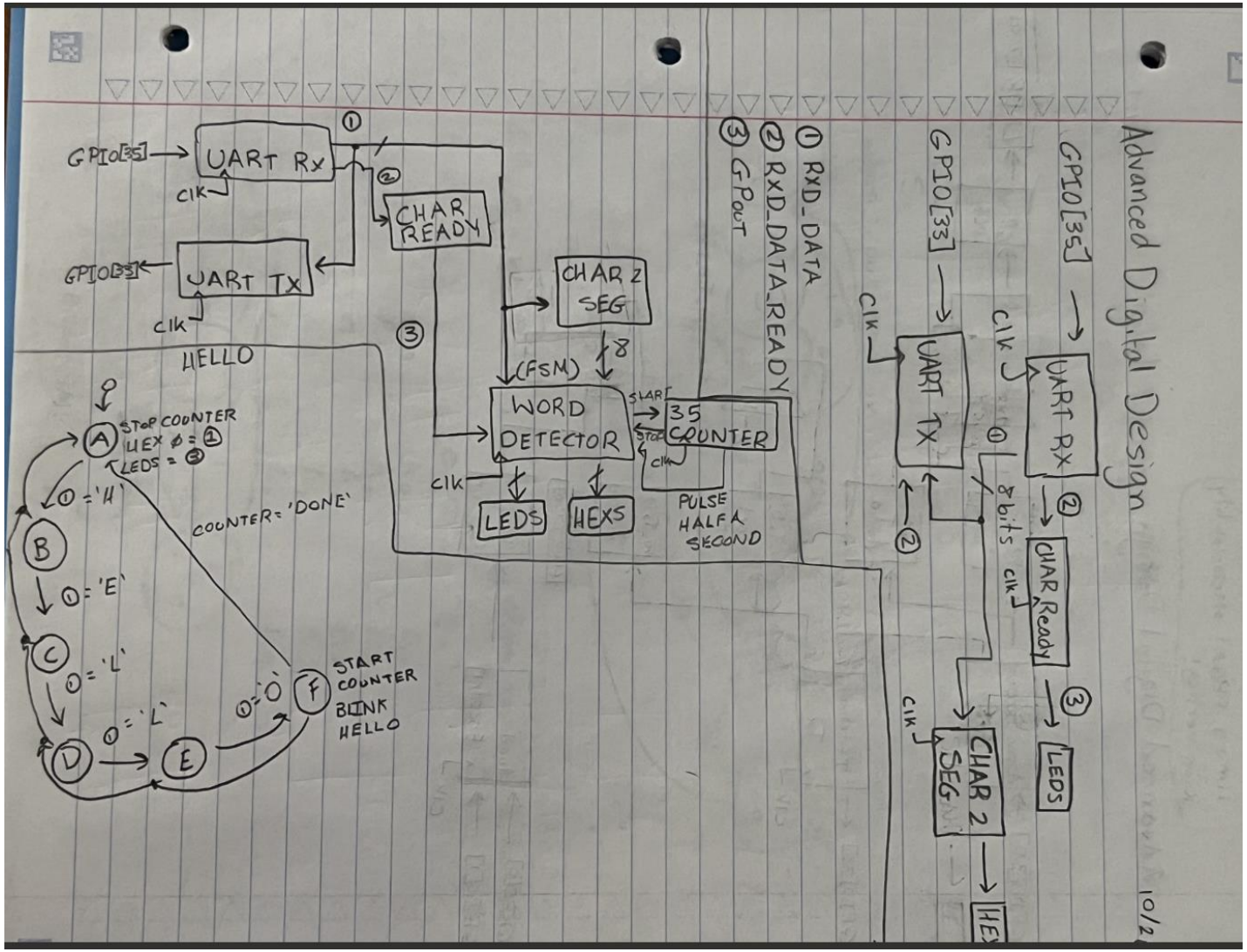
Advanced Digital Design Lab 6

Diego Deoliveira

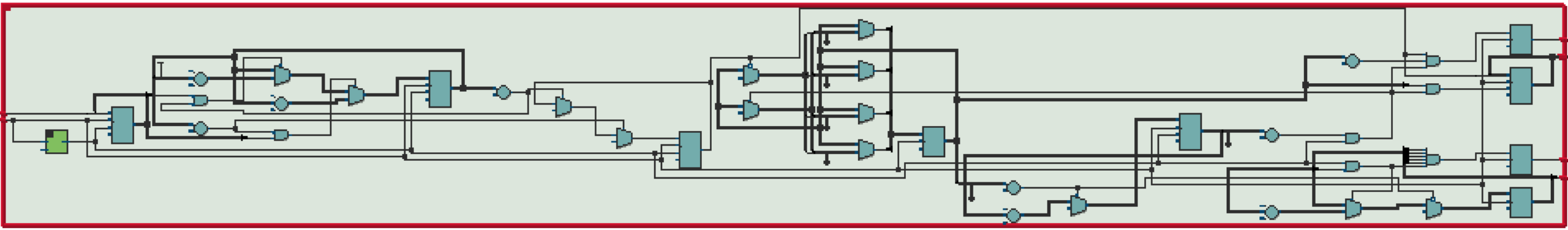
Block and State Diagram of Lab Objective



Async_receiver Explanation

The async_receiver module receives data from another device through a serial connection. In this project, the data enters the FPGA through `GPIO[35]`. The data arrives one bit at a time, so the receiver must collect the bits in the correct order to rebuild the original character.

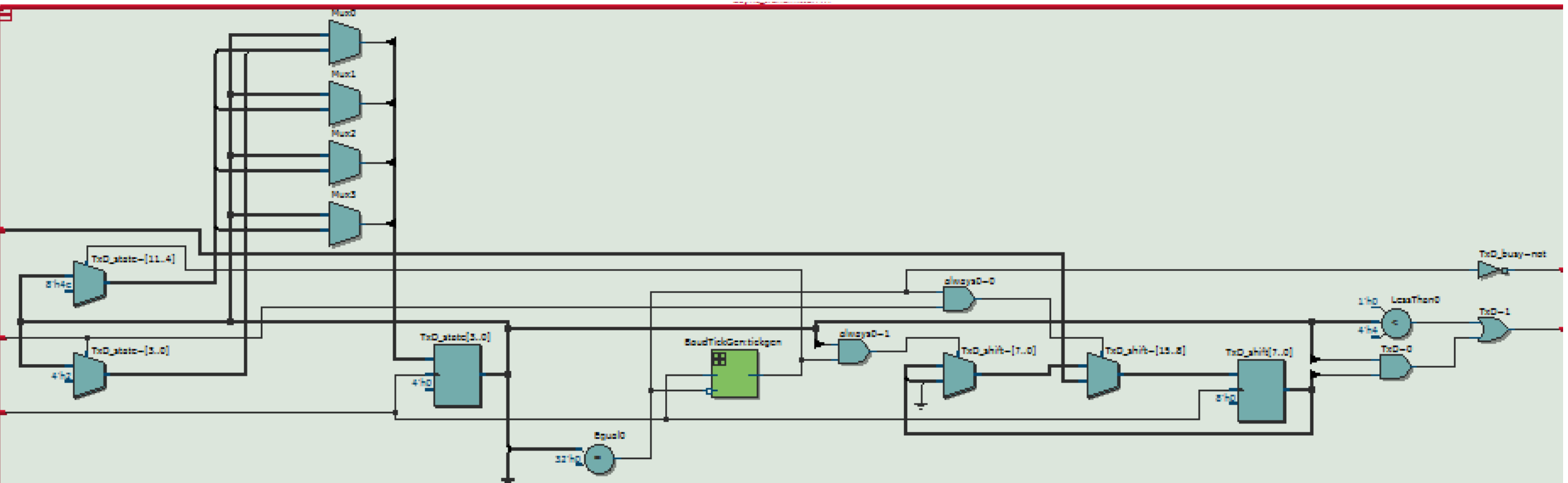
When a character has been received, the module stores it as an 8-bit value and sends a signal called RxD_data_ready. This signal tells the rest of the system that a new character is available. The receiver uses the FPGA's 50 MHz clock and a baud rate of 115200 to help it read the incoming data at the correct times.



Async_transmitter Explanation

The async_transmitter module sends data from the FPGA to another device through a serial connection. In this project, the data leaves through GPIO[33]. Like the receiver, the transmitter sends data one bit at a time.

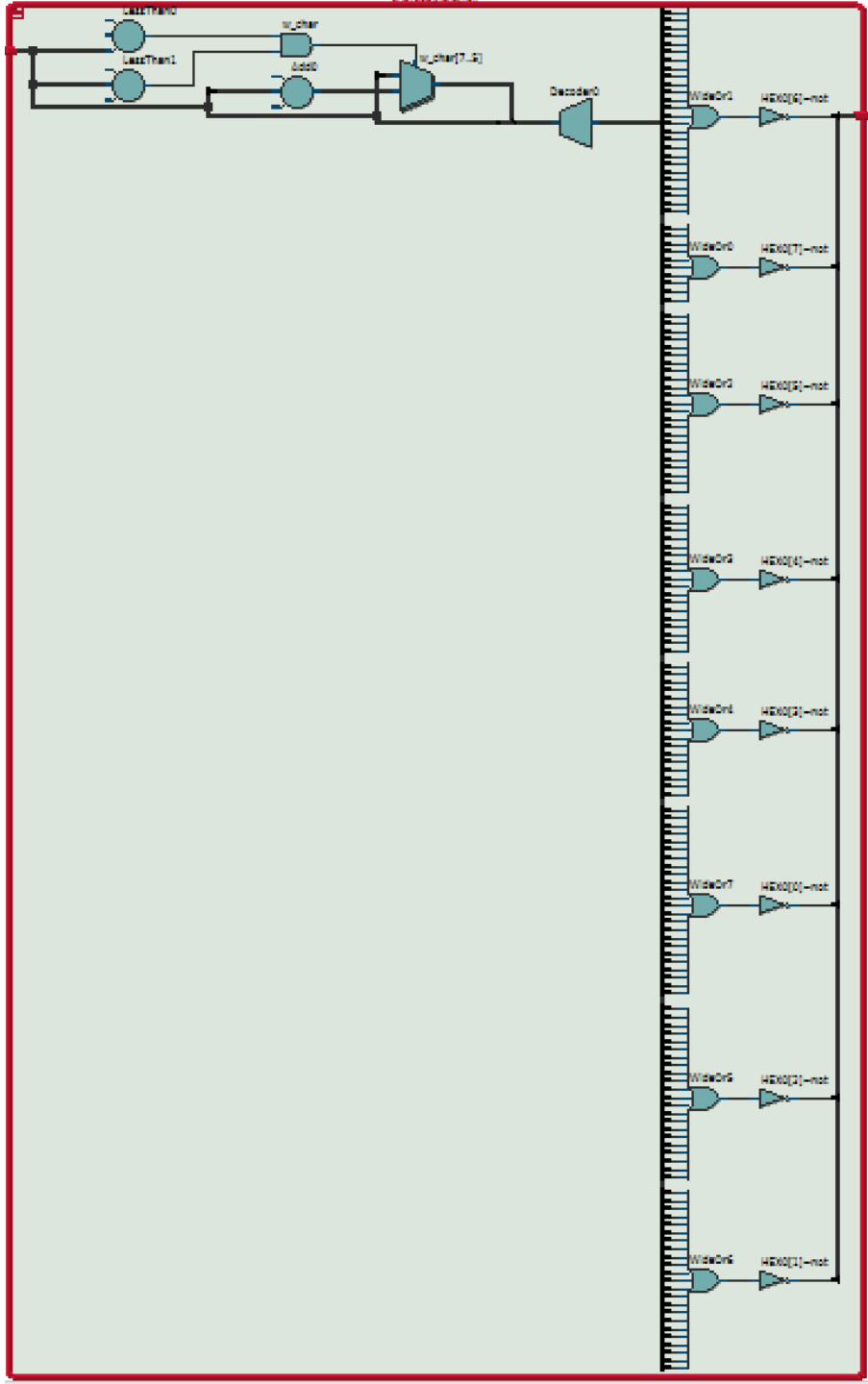
When the transmitter receives a start signal, it sends a start bit, the 8 bits that make up the character, and two stop bits. These bits are sent at a baud rate of 115200. In our design, the transmitter sends the received character back to the device that originally sent it.



Char2seg Explanation

The char2seg module changes a character into the correct signals for the FPGA's seven-segment display, called HEX0. It receives an 8-bit ASCII value, which is a code used to represent a letter or number, and uses that value to choose which display segments should light up.

For example, when the module receives the letter H, it selects the pattern needed to display an H-like shape. The code also converts uppercase letters to lowercase before selecting the pattern, so both forms can use the same display settings. If a character is not included in the code, the display stays off. This module allows the FPGA to show the letters in HELLO on the seven-segment display

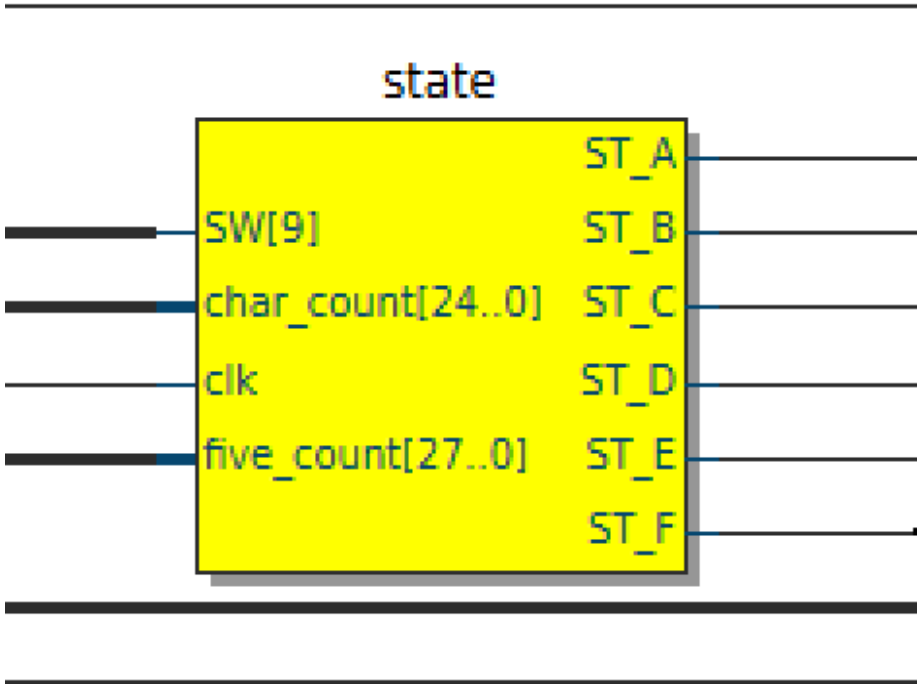


FSM Explanation

The finite state machine (FSM) controls the order of the operations in the system. It uses six states, labeled A, B, C, D, E, and F. Each state represents a different step in the process, and the FSM moves from one state to the next in the order shown in the state diagram.

After reset, the FSM begins at state A and then moves through states B, C, D, and E to display H, E, L, and L. Each letter is shown for approximately half a second. After that, the FSM enters state F, where the display cycles through the letters in HELLO and blinks while the five-second counter runs.

When the five-second counter finishes, the FSM returns to state A and repeats the sequence. Using separate states makes it easier to control which letter is displayed and when the counter starts or stops.



5 second counter Explanation

The five-second counter measures how much time has passed while the FSM is in state F. The FPGA has a 50 MHz clock, which means it produces 50 million clock cycles every second. To measure five seconds, the counter must count 250 million clock cycles.

The counter increases with each clock cycle until it reaches its target. When five seconds have passed, the FSM uses this result to return to state A and restart the sequence. The design also uses smaller counters to control how quickly the letters change and how often the display blinks. This allows the system to control the timing without needing a separate physical timer.

